The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP40015
Subject Title	Intelligent lighting
Credit Value	3
Level	4
Pre-requisite / Co-requisite/ Exclusion	**Exclusion: AP40003**
Objectives	To study the white-light light-emitting diodes (LEDs) technologies for lighting applications. To study the control electronics technologies for lighting and sensing applications. To investigate the potential applications of intelligent lighting.
Intended Learning Outcomes	Upon completion of the subject, students will be able: (a) basic understanding of white-light LEDs technology as well as their electrical and optical properties; (b) basic understanding of the control electronics and devices for the intelligent control of lighting; (c) understanding on the application of artificial intelligence to light control.
Subject Synopsis/ Indicative Syllabus	**White-light Light-Emitting Diodes (LEDs):** Principles of operation of LEDs, heterostructure materials systems, chip design and characteristics. **Control electronics and components of LEDs and sensors**: type of sensors, diodes, transistors, amplifiers, analog to digital conversion; integrated circuits; microprocessor. **Intelligent control systems**: dimmer circuits; networks and buses; cordless control; active architectural and entertainment lighting control. Digital system structures; system design techniques. **Applications of intelligent lighting control**: commercial, industrial and architectural applications. energy management and building control; emergency and security lighting. Nonvisual and visual applications.
Teaching/Learning Methodology	**Lecture:** The course materials will be explained. Concrete examples will be given to illustrate difficult concepts. Students are encouraged to ask questions and participate more in lectures. Assignment sets will be given to assess the learning progress of students. **Tutorial:** Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.
Assessment Methods in Alignment with Intended Learning	

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
		a	b	c

<table>
<tr><td rowspan="2">Outcomes

(Note 4)</td><td colspan="2">
<table>
<tr><td>1. Continuous Assignments</td><td>40%</td><td>√</td><td>√</td><td>√</td></tr>
<tr><td>2. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td></tr>
<tr><td>Total</td><td>100 %</td><td colspan="3"></td></tr>
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</td></tr>
<tr><td colspan="2">Continuous assessment includes assignments and tests which aim at checking the progress of students throughout the course, assisting them in self-monitoring their performances. The examination will be used to assess the knowledge acquired by the students, as well as to determine the extent in which they have achieved the intended learning outcomes.</td></tr>
<tr><td rowspan="5">Student Study Effort Expected</td><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture & Tutorial</td><td>39 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Reading and self-study</td><td>81 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>120 Hrs.</td></tr>
<tr><td>Reading List and References</td><td colspan="2"></td></tr>
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